

Instructions – Applied AI graduate project

You are provided with a 15-year dataset containing monthly measurements of Total Dissolved Solids (TDS), chloride, ammonia, and Biochemical Oxygen Demand (BOD) from a municipal wastewater treatment system. Your task is to use applied AI methods to evaluate long-term trends in wastewater salinity and develop predictive models for future conditions.

Although you may not have prior experience with wastewater treatment, here is essential context: Conventional wastewater treatment relies on microbial communities to remove organic matter (BOD) and convert ammonia to nitrate. High salinity (high TDS) can inhibit these biological processes, reducing treatment efficiency. In regions like Los Angeles, water conservation measures reduce indoor water use, which can unintentionally increase wastewater salinity because the same mass of salts enters the system in a smaller volume of water.

Your objectives

1) Trend Analysis:

Determine whether TDS concentrations have increased over the 15-year period. Use appropriate statistical or machine-learning methods to evaluate trends and quantify the rate of change.

2) Predictive Modeling:

Build a model capable of predicting future TDS concentrations 10 and 20 years from now. You may choose any modeling approach, but justify your selection.

3) Correlation Analysis:

Investigate how increasing salinity may affect biological treatment by analyzing correlations between:

- TDS and ammonia (indicator of nitrification performance)
- TDS and BOD (indicator of organic matter removal)

4) Interpretation in Real-World Context:

Discuss how your findings relate to water conservation practices in Los Angeles and the potential implications for wastewater treatment performance, infrastructure planning, and environmental management. Use the following research paper to guide your discussion: <https://www.nature.com/articles/s41893-020-0529-2>. Please check the links at the end of this document for information on water conservation in Los Angeles.

The dataset needed to complete this assignment is being provided by the instructor. If you want to obtain additional data, click on the link below and enter values according to the following figure.

<https://ciwqs.waterboards.ca.gov/ciwqs/readOnly/CiwqsReportServlet?inCommand=reset&reportName=esmrAnalytical>

eSMR Analytical Report

Region:

Region 1 - North Coast
Region 2 - San Francisco Bay
Region 3 - Central Coast
Region 4 - Los Angeles

Note: Hold "Ctrl" while clicking to select multiple values

Report Document Type:

Any

Program:

Any

Parameter:

Toluene-d8
Total Aroclors
Total Coliform
Total Dissolved Solids (TDS)

WDID:

Sample Date Range: Start:

07/01/2010

 (MM/DD/YYYY) End:

06/01/2026

 (MM/DD/YYYY)

Order Number:

Facility:

Los Angeles-Glendale WWRP

Select Facility

Clear Facility

Party:

Select Party

County:

Kings
Lake
Lassen
Los Angeles

Monitoring Location Type:

Any

Record Type:

All

Run Report

Useful links:

https://sanitation.lacity.gov/san/faces/home/portal/s-lsh-wwd/s-lsh-wwd-cw/s-lsh-wwd-cw-p/s-lsh-wwd-cw-p-lagwrp?_adf.ctrl-state=trf7gmcxr_1&_afLoop=1106679206338114&_afWindowMode=0&_afWindowId=1byys0qoef#!%40%40%3F_afWindowId%3D1byys0qoef%26_afLoop%3D1106679206338114%26_afWindowMode%3D0%26_adf.ctrl-state%3Dtrf7gmcxr_5

https://sanitation.lacity.gov/san/faces/home/portal/s-lsh-es/s-lsh-es-owla/s-lsh-es-owla-r?_adf.ctrl-state=trf7gmcxr_5&_afLoop=1106789389116833#!

<https://www.ladwp.com/who-we-are/water-system/water-conservation>